Kreisel High Voltage Battery

Can Trace Logging

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1 Version history

- V1 : Initial version 25/10/2020
- V2 : Removed post processing. 23/2/2021
- V3 : Added file content checks. Added logging of sensor can trace. 30/3/2021

2 Equipment required

- Laptop computer with Ixxat drivers and canAnalyser3Mini software tool installed.
- Ixxat USB to CANbus interface module.
- Battery control connector interface harness, or pair of TE Ampseal female contacts 770854-3 , along with 0.5 to 1mm² wire. (to connect to sensor can).
- Delphi (Aptiv) 150 male 2 way connector (connector body Delphi part number 12162000 + male contact 0.5-1.0mm² Delphi part number 12045773).

3 No high voltage handling required

- Despite the battery being a high voltage battery (324V nominal voltage, 378V maximum cut-off voltage), the present diagnostic does not require to work on the battery high voltage

part.

The diagnostic is done through one of the two communications port of the battery, and as such, only requires to unplug and replug the low voltage (12V supplied) control connector of the battery.

4 Summary of the work to be done

- The high voltage battery has two CAN buses : the control CAN bus, and the sensor CAN bus.
  - Only the control CAN bus is used in normal use of the boat.
  - The sensor CAN bus is a battery factory diagnostic bus, and as such is not wired on the boat harness.
- For an extended battery diagnostic by the battery manufacturer, both CAN buses have to be recorded from boat switch-on upto the occurrence of the battery error.
- Since the USB to CANbus interface module only has one channel, the boat switch-on sequence will be done twice, with only one CANbus being recorded at a time.
- Recorded files will then have to be sent to Iguana, for transmission to the battery manufacturer for analysis.

5 Recording a CANbus trace

.....*5.1 Iguana CANbus networks, and battery CANbus networks*

CAN buses on the Iguana harness

- The Iguana Commuter master controller (MC2) has three CANbus networks :
  - Main system CANbus, labelled « Main », also linked to a Bluetooth module (Parker G11) for remote connection through the IQANgo smartphone application.
  - Electric drive components CANbus, labelled « Elec », which links the Master Controller (Parker MC42) to the :
    - track motors inverters (Sevcon Gen 4 Size 8)
    - hydraulic pump power inverter (Sevcon HVLP 20)
    - high voltage battery charger (Brusa NLG513 or Elcon HK-J-H440-10)
  - Traction battery CANbus, labelled « BatCan » or « Aux ». That network is the only battery CANbus (control CANbus) which is wired to the Iguana harness.

    The other battery CANbus (sensor CANbus) is not wired to the Iguana harness.

CAN buses of the Kreisel battery

- The Kreisel battery has two CANbus networks, which both have to be recorded for the present diagnostic, but only one is wired to the Iguana harness :
  - Control CAN bus, wired to the Iguana traction battery CANbus, labelled « BatCan » or « Aux » (see above).

- Sensor CAN bus (pins 14 for Can High and 22 for Can low, of the battery TE Ampseal 23 pin control signals connector), not wired to the Iguana harness. It therefore has to be wired to a wire pair that will connect to the USB to CANbus interface connector for the purpose of this diagnostic.

.....*5.2* *Setting up IXXAT USB to CAN interface module*

So as to interface a laptop computer to the battery CANbus network to be monitored.

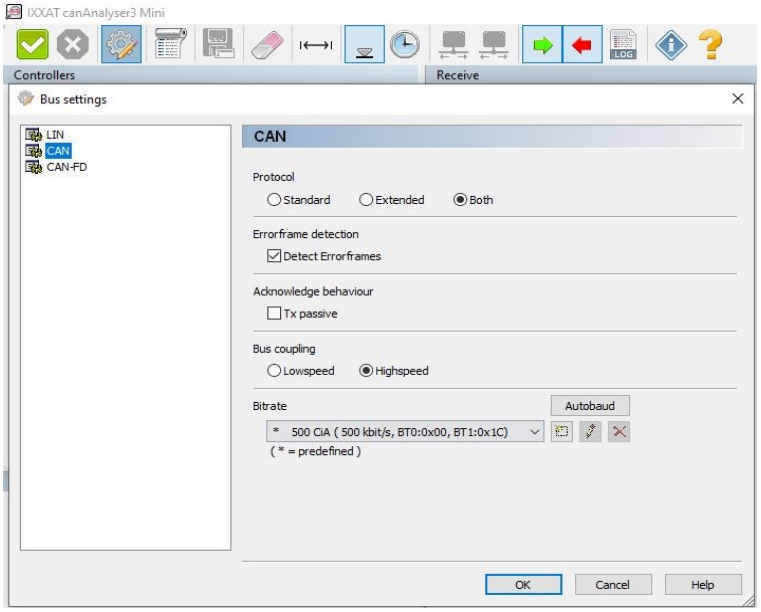
- A CANbus to USB interface module similar to the one shown on the photo hereunder is necessary.
  - On Commuters, it is supplied within the ex-factory boat kit.
  - On Originals, it is supplied as an optional servicing accessory.



- Have the IXXAT interface module drivers installed onto a laptop computer : Ixxat VCI Setup 4.0.925.0.exe installation, followed by installation of the add-on VCI V2 API-Addon (VCI2 on VCI4)_2_31_133_0.exe.

  That will also install the canAnalyser3Mini application on the laptop.

- Launch the canAnalyser3Mini application on the laptop and set its settings (gearwheel icon) as shown hereunder. That application will allow to read and record CANbus communications from the battery.



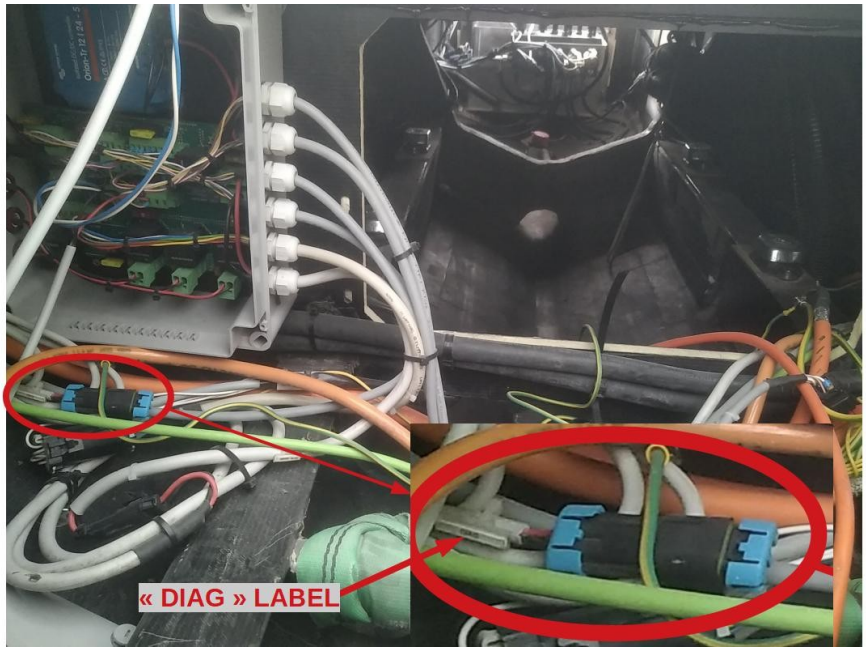
- Then, before launching the recording (white tick in the green box), set the recording files and format by clicking on the paper roll icon. Choose a file location, file name (without extension) and set « file type » to all, so that both .csv and .tr0 are recorded. Tr0 file types will be required for some decoding tools at the manufacturer's level.

.....5.3 *CAN trace of the HV battery control CANbus*

Connection of the USB to CANbus module to the Iguana system

On Iguana Original (center console Iguana)

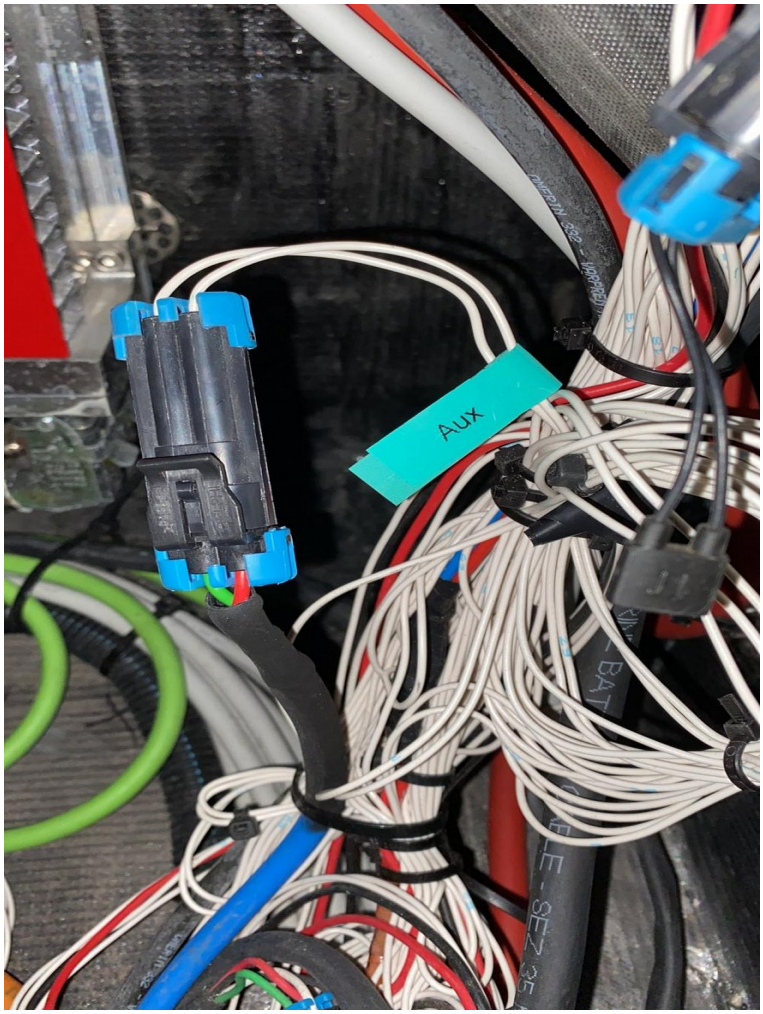
- Ixxat USB to CAN interface module has to be connected to the Iguana system into a diagnostic connector labelled « Diag » that can be found in the front bilge (under the front deck hatch), next to the grey electric box placed on the starboard side of the compartment (attached "DiagConnector" photo herewith).



Remove the "Diag" connector cap, and connect the Diag connector to the mating connector on the "USB to CAN" interface module harness.

On Iguana Commuters (cabin Iguana)

- Ixxat USB to CAN interface module has to be connected to the Iguana system in series with the CANbus labelled « BatCan » or « Aux », which can be found underneath the floor hatch of the recess to portside of the cabin steps. The CANbus is a twisted pair of blue and white, or white and white wires. The interface module should be plugged in series where a connector matting is found on this bus harness.



CANbus trace recording routine common to all Iguana models

- Plug the module USB port into the laptop.
- Turn on the Iguana system from the key switch on the center console, and then :
  - if on an Iguana Original : press on F1 (start).
  - If on an Iguana Commuter : click on the Tracks softkey of the touch panel display.
- On the laptop application, click on the "start all communications" icon (white tick in green box).
- You should then see lines of digital messages scroll down the receive window of canAnalyser3 on the laptop screen.

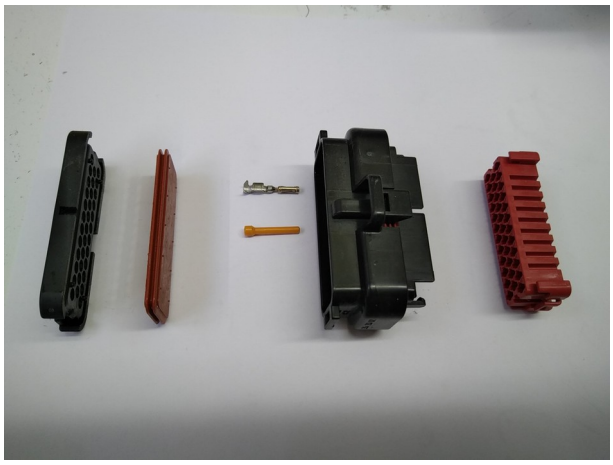
  If so, the setup is ready for recording the battery communication messages. We can now proceed with the actual diagnostic recording.
- Turn off the Iguana system from the key switch on the center console.
- Click on the "clear receive window" icon (rubber icon) on the laptop application, then on the

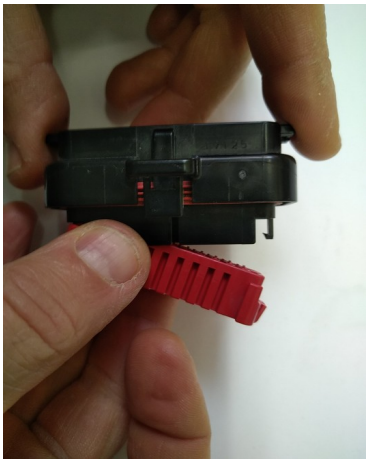
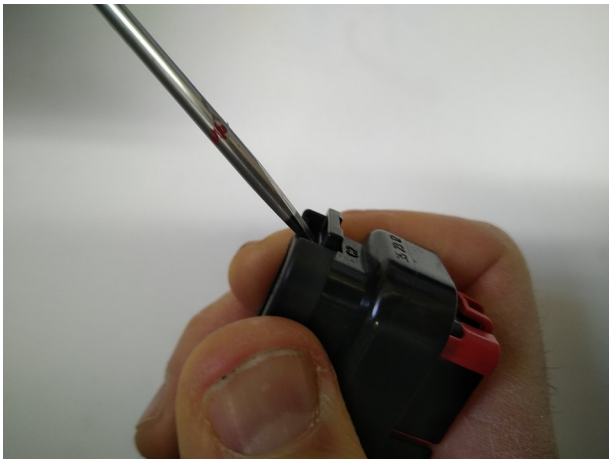
"inline write to file" icon (paper roll icon). Choose a directory where to store the file, file name (without extension) and set « file type » to all, so that both .csv and .tr0 are recorded. Tr0 file types will be required for some decoding tools at the manufacturer's level.

- Turn on the Iguana system again from the key switch on the center console, and then (if switching the tracks on is necessary to have the battery error appear) :
  - if on an Iguana Original : press on F1 (start).
  - If on an Iguana Commuter : click on the Tracks softkey of the touch panel display.
- Once the battery error message has appeared on the Iguana display, stop the recording on the laptop application.
- The recroding is now complete for the control CANbus. Retrieve both the .tr0 and .csv recorded file (IXXAT_canAnalyser3_MiniLog_20-10-17_.....tr0).
- Check that both .tr0 and .csv file formats have been stored on your laptop (the "tr0" file is mandatory) and **check they are filled-up with data.** They should be over 1Mbyte in size for one minute of recording (about 1.5Mbytes/mn for the .tr0, 6Mbytes/mn for the .csv).
- Open the .csv file with a text editor, and check :
  - at the top of the file that bitrate is mentioned to be 500 kbits/s.
  - In the file text that there is no occurrence of the word "error". If so, retry the recording or contact Iguana with the file as reference. Error will occur if the battery reboots during the recording, in which case the presence of the word « error » makes sense and is acceptable.
- Now that the battery control CANbus trace is recorded, unplug the USB to CANbus interface module from the Iguana harness. Replug the Iguana harness mating connectors together so as to put the harness back to its initial configuration.

.....*5.4 Preparing the connection to the battery sensor CAN pins*

- As said above, since the sensor CAN bus is a battery factory diagnostic bus, and as such is not wired on the boat harness, connections to it have to be made in the field at this stage.
- Either a Kreisel diagnostic harness has been provided by Iguana, or the connection can be made using :
  - On the battery side : TE Ampseal female contacts 770854-3 (for 0.5 to 1.5mm² wire).
    - Sensor Can High on pin 14 of the battery TE Ampseal 23 pin control signals connector.
    - Sensor Can Low on pin 22 of the battery TE Ampseal 23 pin control signals connector.



- On the USB to CAN bus module side, a Delphi (Aptiv) 150 male 2 way connector (connector body Delphi part number 12162000 + male contact 0.5-1.0mm² Delphi part number 12045773).
  - Sensor Can High in Delphi 150 connector cavity A.
  - Sensor Can Low in Delphi 150 connector cavity B.

.....5.5 *CAN trace of the HV battery sensor CANbus*

- Now that the battery control CANbus trace is recorded, a recording using exactly the same process has to be performed on the battery sensor CANbus.
- After having added Ampseal female contacts to the battery female connector, and a Delphi 150 connector to the loose end of the wire pair, plug the mating connector of the USB to CANbus interface module to the battery sensor CANbus wire pair.
- Proceed to the sensor CANbus trace recording exactly in the same way as described in the previous paragraph « CANbus trace recording routine common to all Iguana models ».
- Check that both .tr0 and .csv file formats have been stored on your laptop (the “tr0” file is mandatory) and **check they are filled-up with data.** They should be over 1Mbyte in size for one minute of recording (about 1.5Mbytes/mn for the .tr0, 6Mbytes/mn for the .csv).
- Open the .csv file with a text editor, and check :
  - at the top of the file that bitrate is mentioned to be 500 kbits/s.

- In the file text that there is no occurrence of the word “error”. If so, retry the recording or contact Iguana with the file as reference. Error will occur if the battery reboots during the recording, in which case the presence of the word « error » makes sense and is acceptable.

- Disconnect the USB to CANbus interface module, remove or plug the Sensor CANbus wire pair or Kreisel diagnostic interface harness, so as to put the boat harness back to a standard use configuration.
- Please kindly email all four files (two for Control CANbus trace, and two for Sensor CANbus trace) over to Iguana after sales for analysis with the battery manufacturer.

6 Post-processing files (for Iguana internal use)

- So as to convert them to .asc files. See dedicated instructions sheet.